

EXP21: Prolog Program to Implement Towers of Hanoi

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Aim:

To develop a Prolog program that solves the Towers of Hanoi problem for N disks and displays the complete sequence of moves required.

Algorithm:

Step 1: Start the program.

Step 2: Specify the number of disks and the source, destination, and auxiliary rods.

Step 3: Check whether there are any disks to move.

Step 4: If the number of disks is greater than zero, move the top N-1 disks from the source rod to the auxiliary rod.

Step 5: Move the remaining largest disk from the source rod to the destination rod.

Step 6: Move the N-1 disks from the auxiliary rod to the destination rod.

Step 7: Repeat the process recursively until all disks are transferred.

Step 8: Display each move performed.

Step 9: Stop the program.

Source Code:

```
% Program to implement Towers of Hanoi
```

```
hanoi(0, _, _, _) :- !.
```

```
hanoi(N, From, To, Via) :-
```

```
    N > 0,
```

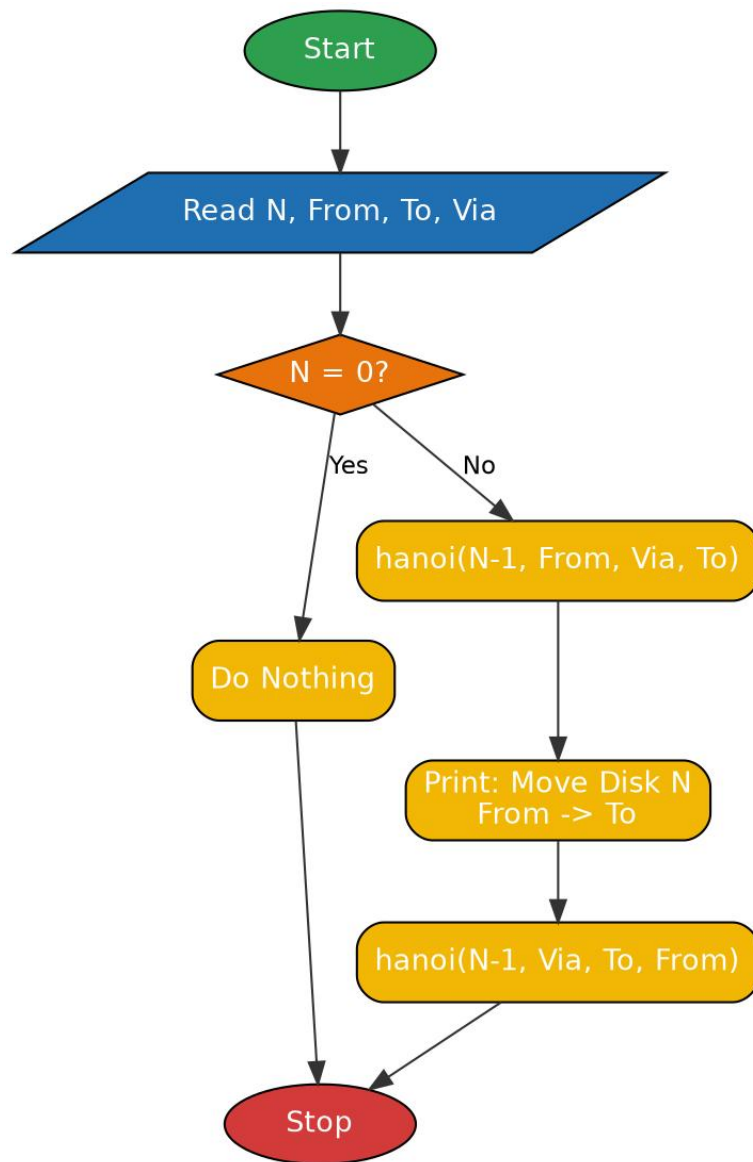
```
    N1 is N - 1,
```

```
    hanoi(N1, From, Via, To),
```

```
    format("Move disk ~w from ~w to ~w~n", [N, From, To]),
```

```
    hanoi(N1, Via, To, From).
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [hanoi].
true.

?- hanoi(3, left, right, middle).
Move disk 1 from left to right
Move disk 2 from left to middle
Move disk 1 from right to middle
Move disk 3 from left to right
Move disk 1 from middle to left
Move disk 2 from middle to right
Move disk 1 from left to right
true.

?- hanoi(4, left, right, middle).
Move disk 1 from left to middle
Move disk 2 from left to right
Move disk 1 from middle to right
Move disk 3 from left to middle
Move disk 1 from right to left
Move disk 2 from right to middle
Move disk 1 from left to middle
Move disk 4 from left to right
Move disk 1 from middle to right
Move disk 2 from middle to left
Move disk 1 from right to left
Move disk 3 from middle to right
Move disk 1 from left to middle
Move disk 2 from left to right
Move disk 1 from middle to right
true.
```

Result:

Thus the Prolog program to implement the Towers of Hanoi problem was written, executed successfully and the output was verified.